

LATTICES IN CRYPTOSYSTEMS

DRP: INTRODUCTION TO CRYPTOGRAPHY

Andrew Hwang

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1 Introduction to Lattices

Definition 1.1. Lattice

Let $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n \in \mathbb{R}^m$ be linearly independent vectors. The **lattice** generated by these vectors is the set

$$\mathcal{L} = \mathcal{L}(\mathbf{v}_1, \dots, \mathbf{v}_n) = \left\{ \sum_{i=1}^n a_i \mathbf{v}_i \mid a_i \in \mathbb{Z} \right\}.$$

The vectors $\mathbf{v}_1, \dots, \mathbf{v}_n$ form a **basis** for $\mathcal{L}$, and n is the **rank** (or dimension) of the lattice.

A lattice is a discrete additive subgroup of $\mathbb{R}^m$. Unlike a vector space, only discrete *integer* linear combinations of the basis vectors are allowed.

Definition 1.2. Fundamental Domain

Given a lattice $\mathcal{L}$ with basis $\mathbf{v}_1, \dots, \mathbf{v}_n$, the **fundamental domain** (or fundamental parallelepiped) is

$$\mathcal{F} = \left\{ \sum_{i=1}^n t_i \mathbf{v}_i \mid 0 \leq t_i < 1 \right\}.$$

The volume of $\mathcal{F}$ is $\det(\mathcal{L}) = |\det(\mathbf{V})|$, where $\mathbf{V}$ is the matrix whose rows are $\mathbf{v}_1, \dots, \mathbf{v}_n$. This quantity is independent of the choice of basis.

1.1 Two-Dimensional Lattices

In two dimensions, a lattice $\mathcal{L} \subset \mathbb{R}^2$ is generated by two linearly independent vectors $\mathbf{v}_1, \mathbf{v}_2$. The lattice points form a regular grid of parallelograms tiling the plane.

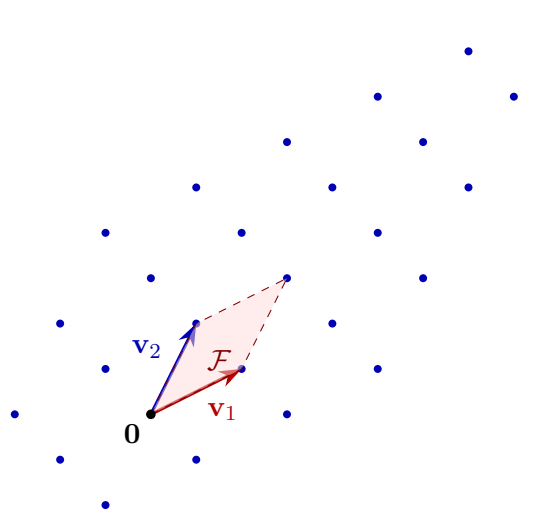


Figure 1: An example of a two-dimensional lattice

2 The Congruential Public Key Cryptosystem

We now describe a public key cryptosystem based on modular arithmetic. This system illustrates how lattice methods can be used to attack cryptographic schemes.

2.1 Key Creation (Alice)

Alice performs the following steps:

1. Choose a large integer modulus q .
2. Choose secret integers f and g satisfying:
 - $f < \sqrt{q/2}$,
 - $\sqrt{q/4} < g < \sqrt{q/2}$,
 - $\gcd(f, g) = 1$.
3. Compute $h \equiv f^{-1}g \pmod{q}$.
4. Publish the **public key** (q, h) .
5. Keep f and g as her **private key**.

Remark 2.1. Size Constraints

The constraints on f and g are critical. Both are roughly of size $\sqrt{q}$, which is much smaller than q . The public value $h \equiv f^{-1}g \pmod{q}$ looks like a random element of $\mathbb{Z}/q\mathbb{Z}$, so recovering the small values f and g from h and q is, at first glance, a hard problem. However, as we will see, it is precisely this structure that a lattice attack exploits.

2.2 Encryption (Bob)

Bob wants to send a plaintext message m to Alice. He proceeds as follows:

1. Choose a plaintext m with $m < \sqrt{q/2}$.
2. Choose a random integer r (the ephemeral key).
3. Using Alice's public key (q, h) , compute the ciphertext:

$$e \equiv r \cdot h + m \pmod{q}.$$

4. Send the ciphertext e to Alice.

2.3 Decryption (Alice)

Upon receiving e , Alice uses her private key (f, g) :

1. Compute $a \equiv f \cdot e \pmod{q}$ with $0 < a < q$.
2. Compute $b \equiv f^{-1}a \pmod{g}$ with $0 < b < g$.
3. Then b is the plaintext m .

Since $h \equiv f^{-1}g \pmod{q}$, we have

$$a \equiv f \cdot e \equiv f(rh + m) \equiv f \cdot r \cdot f^{-1}g + fm \equiv rg + fm \pmod{q}.$$

When the parameters are small enough, the value $rg + fm$ does not wrap around modulo q , so $a = rg + fm$ exactly. Then

$$b \equiv f^{-1}a \equiv f^{-1}(rg + fm) \equiv f^{-1} \cdot rg + m \equiv 0 + m \equiv m \pmod{g},$$

since $g \mid rg$. Hence $b = m$ (provided $m < g$).

3 Breaking the System with a 2D Lattice

Let Eve be the eavesdropper. Eve knows the public key (q, h) and wants to recover Alice's private key (f, g) . She knows that

$$h \equiv f^{-1}g \pmod{q} \iff f \cdot h \equiv g \pmod{q} \iff f \cdot h - g = k \cdot q \text{ for some } k \in \mathbb{Z}.$$

3.1 Constructing the Lattice

Eve constructs the following two-dimensional lattice. Define basis vectors

$$\mathbf{v}_1 = \begin{pmatrix} 1 \\ h \end{pmatrix}, \quad \mathbf{v}_2 = \begin{pmatrix} 0 \\ q \end{pmatrix}.$$

The lattice $\mathcal{L} = \mathcal{L}(\mathbf{v}_1, \mathbf{v}_2)$ consists of all integer combinations

$$a\mathbf{v}_1 + b\mathbf{v}_2 = \begin{pmatrix} a \\ ah + bq \end{pmatrix}, \quad a, b \in \mathbb{Z}.$$

Remark 3.1. The key observation

Notice that if Eve sets $a = f$ and $b = -k$ (where $fh - g = kq$), then

$$f\mathbf{v}_1 + (-k)\mathbf{v}_2 = \begin{pmatrix} f \\ fh - kq \end{pmatrix} = \begin{pmatrix} f \\ g \end{pmatrix}.$$

So the vector $\mathbf{t} = (f, g)$ is a lattice point in $\mathcal{L}$!

3.2 Why (f, g) is a Short Vector

The determinant of the lattice is

$$\det(\mathcal{L}) = \det \begin{pmatrix} 1 & h \\ 0 & q \end{pmatrix} = q.$$

Hence, by the Gaussian heuristic, the shortest vector in a two-dimensional lattice of determinant q has length approximately $\sqrt{q}$. Now recall the size constraints

$$f < \sqrt{q/2}, \quad g < \sqrt{q/2}.$$

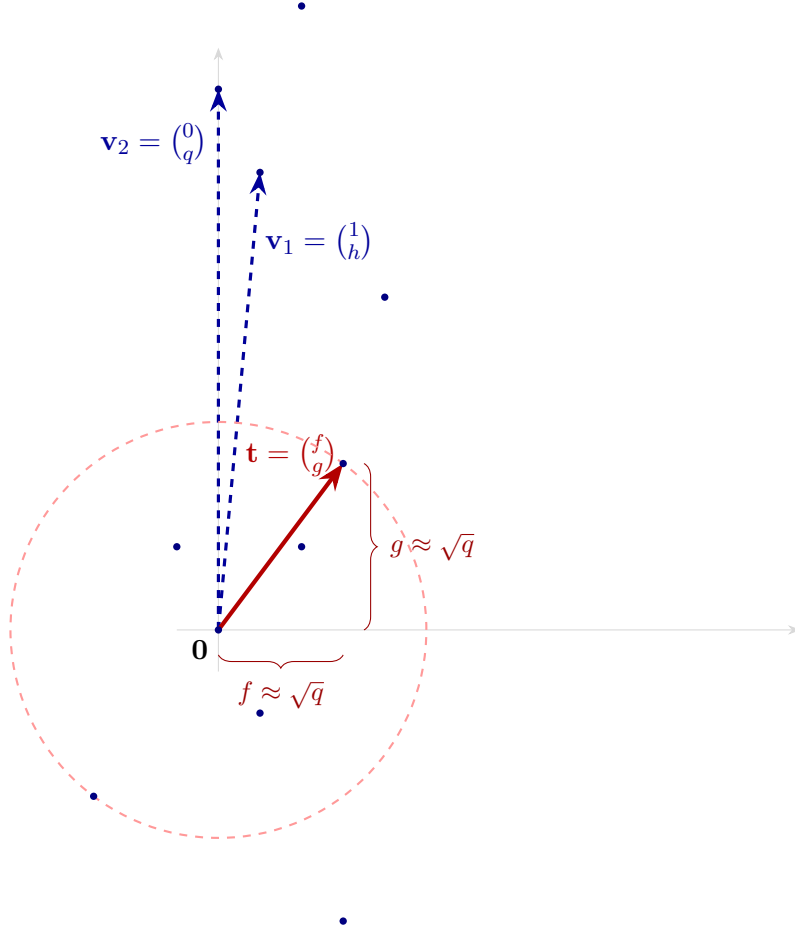
So, the vector (f, g) has norm

$$\|(f, g)\| = \sqrt{f^2 + g^2} < \sqrt{q/2 + q/2} = \sqrt{q}.$$

This means (f, g) is *exceptionally short* for a vector in a lattice of determinant q ; it is roughly the size of the shortest vector predicted by the Gaussian heuristic. The given basis vectors $\mathbf{v}_1 = (1, h)$ and $\mathbf{v}_2 = (0, q)$ are very long (of norm approximately $h \approx q$ and q respectively), but the lattice contains the much shorter vector (f, g) .

3.3 The Lattice Geometry

The following diagram illustrates the situation. The original basis vectors $\mathbf{v}_1$ and $\mathbf{v}_2$ are long and nearly parallel (from the perspective of the lattice determinant), while the target vector (f, g) is short.



3.4 Finding the Short Vector

In dimension 2, Eve can use **Gaussian lattice reduction** to find the shortest vector efficiently. The algorithm works as follows:

Gaussian Lattice Reduction (Dimension 2)

Input: Basis $\{\mathbf{v}_1, \mathbf{v}_2\}$ with $\|\mathbf{v}_1\| \leq \|\mathbf{v}_2\|$.

Output: Reduced basis with $\|\mathbf{v}_1\| \leq \|\mathbf{v}_2\|$ and $|\langle \mathbf{v}_1, \mathbf{v}_2 \rangle| \leq \frac{1}{2} \|\mathbf{v}_1\|^2$.

1. If $\|\mathbf{v}_1\| > \|\mathbf{v}_2\|$, swap $\mathbf{v}_1 \leftrightarrow \mathbf{v}_2$.
2. Compute $\mu = \left\lfloor \frac{\langle \mathbf{v}_1, \mathbf{v}_2 \rangle}{\langle \mathbf{v}_1, \mathbf{v}_1 \rangle} \right\rfloor$ (nearest integer).
3. Replace $\mathbf{v}_2 \leftarrow \mathbf{v}_2 - \mu \mathbf{v}_1$.
4. If $\|\mathbf{v}_1\| \leq \|\mathbf{v}_2\|$, **stop**. Else, go to step 1.

When Eve runs this algorithm on the basis $\{(1, h), (0, q)\}$, it will, in a small number of

iterations, produce a short basis for $\mathcal{L}$. Since (f, g) is approximately the shortest vector in the lattice, the reduced basis will reveal (f, g) (or $\pm(f, g)$), thereby recovering Alice's private key.

3.5 Example

Example 3.1. A Small Instance

Let $q = 541$ and suppose Alice chooses $f = 13$, $g = 15$ (with $\gcd(13, 15) = 1$). Note that $\sqrt{q/2} \approx 16.4$, so both f and g satisfy the size constraints. Alice computes:

$$h \equiv f^{-1}g \equiv 13^{-1} \cdot 15 \pmod{541}.$$

Since $13 \cdot 125 = 1625 = 3 \cdot 541 + 2$, we need $13^{-1} \pmod{541}$. One finds $13^{-1} \equiv 375 \pmod{541}$ (since $13 \times 375 = 4875 = 9 \times 541 + 6$... let us just compute it properly: $13 \cdot 125 = 1625 \equiv 1625 - 3 \cdot 541 = 1625 - 1623 = 2$, so $13 \cdot 125 \equiv 2$, thus $13 \cdot 250 \equiv 4$, not quite).

Let us simply set up the system. Suppose after correct computation we find $h = 283$. Then Alice's public key is $(q, h) = (541, 283)$.

Eve's attack. Eve constructs the lattice with basis:

$$\mathbf{v}_1 = \begin{pmatrix} 1 \\ 283 \end{pmatrix}, \quad \mathbf{v}_2 = \begin{pmatrix} 0 \\ 541 \end{pmatrix}.$$

She applies Gaussian reduction:

1. $\|\mathbf{v}_1\|^2 = 1 + 283^2 = 80090$, $\|\mathbf{v}_2\|^2 = 541^2 = 292681$, so $\|\mathbf{v}_1\| < \|\mathbf{v}_2\|$. Good.
2. $\mu = \lfloor \frac{283 \cdot 541}{80090} \rfloor = \lfloor \frac{153103}{80090} \rfloor = \lfloor 1.911 \dots \rfloor = 2$.
3. $\mathbf{v}_2 \leftarrow (0, 541) - 2(1, 283) = (-2, -25)$.
4. Now $\|\mathbf{v}_2\|^2 = 4 + 625 = 629 < 80090 = \|\mathbf{v}_1\|^2$. Swap: $\mathbf{v}_1 = (-2, -25)$, $\mathbf{v}_2 = (1, 283)$.
5. After a few more iterations, the algorithm terminates with a short basis. The shortest vector found will be $\pm(f, g) = \pm(13, 15)$ (or a vector closely related to it).

Eve has recovered the private key!

References

- [1] J. Hoffstein, J. Pipher, and J.H. Silverman, *An Introduction to Mathematical Cryptography*, 2nd ed., Undergraduate Texts in Mathematics, Springer, New York, 2014.